

Supervised Learning, COMP0078
Sample Examination Paper 1
Suitable for Cohorts: 25/26

Answer ALL EIGHT (8) questions.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

Time Allowed: 3 Hours

1. (k-Nearest Neighbors and Bayes Error)

- (a) Let $\mathcal{Y} = \{1, \dots, C\}$ be a set of C classes. Given distribution P over $\mathcal{X} \times \mathcal{Y}$, define the Bayes classifier f^* and the Bayes error rate. [3 marks]
- (b) Consider the 1-NN classifier. As the number of training samples $m \rightarrow \infty$, show that the expected error rate of 1-NN at a point $\mathbf{x}$ is given by:

$$\sum_{c=1}^C P(c|\mathbf{x})[1 - P(c|\mathbf{x})]$$

Explain the key assumption required for this result to hold. [5 marks]

- (c) Prove that the 1-NN error rate is at most twice the Bayes error rate. That is, show:

$$\sum_{c=1}^C P(c|\mathbf{x})[1 - P(c|\mathbf{x})] \leq 2 \left[1 - \max_{c \in \{1, \dots, C\}} P(c|\mathbf{x}) \right]$$

[7 marks]

- (d) State conditions on $k(m)$ such that the $k(m)$ -NN classifier achieves the Bayes error rate as $m \rightarrow \infty$. Briefly explain the curse of dimensionality in the context of k -NN. [5 marks]

2. (Kernels and Dual Representations)

Consider ridge regression with squared loss:

$$\min_{\mathbf{w} \in \mathbb{R}^n} \left\{ \frac{1}{m} \sum_{i=1}^m (\mathbf{w}^\top \mathbf{x}_i - y_i)^2 + \lambda \|\mathbf{w}\|^2 \right\}$$

- (a) Let $X \in \mathbb{R}^{m \times n}$ denote the data matrix and $\mathbf{y} \in \mathbb{R}^m$ the output vector. Derive the closed-form solution $\mathbf{w}^*$ for the primal problem. [3 marks]

- (b) Using the representer theorem, show that the solution can be written as $\mathbf{w}^* = X^\top \boldsymbol{\alpha}$ for some $\boldsymbol{\alpha} \in \mathbb{R}^m$. Derive the explicit form of $\boldsymbol{\alpha}$. [6 marks]
- (c) For a new test point $\mathbf{x}$, express the prediction $\mathbf{w}^{*\top} \mathbf{x}$ in terms of inner products $\langle \mathbf{x}_i, \mathbf{x} \rangle$ only. Explain how this allows the use of kernel functions. [4 marks]
- (d) Compare the computational complexity of solving the primal versus dual problem. Under what conditions is each approach preferable? [4 marks]
- (e) Consider the polynomial kernel $k(\mathbf{x}, \mathbf{z}) = (1 + \mathbf{x}^\top \mathbf{z})^2$ for $\mathbf{x}, \mathbf{z} \in \mathbb{R}^2$. Write out the explicit feature map $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^N$ such that $k(\mathbf{x}, \mathbf{z}) = \phi(\mathbf{x})^\top \phi(\mathbf{z})$. [3 marks]

3. (Support Vector Machines)

- (a) Define the geometric margin of a hyperplane $H_{\mathbf{w},b} = \{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b = 0\}$ with respect to a point $\mathbf{x}$. State the canonical parameterization of a separating hyperplane. [3 marks]
- (b) Write down the primal optimization problem for the hard-margin SVM. Formulate the Lagrangian and derive the dual problem by computing the KKT conditions. [8 marks]
- (c) For the soft-margin SVM with slack variables ξ_i , the primal problem is:

$$\min_{\mathbf{w}, b, \xi} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^m \xi_i \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Show that this is equivalent to minimizing:

$$\sum_{i=1}^m \max(0, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)) + \lambda \|\mathbf{w}\|^2$$

Identify the loss function and express λ in terms of C . [6 marks]

- (d) Explain the geometric interpretation of the KKT conditions for the soft-margin case. Specifically, describe what happens when $\alpha_i = 0$, $0 < \alpha_i < C$, and $\alpha_i = C$. [3 marks]

4. (Decision Trees and Ensemble Methods)

- (a) In the CART algorithm for regression trees, given a splitting variable j and split point s , we define regions $R_1(j, s) = \{\mathbf{x} : x_j \leq s\}$ and $R_2(j, s) = \{\mathbf{x} : x_j > s\}$. Write down the optimization problem for finding the optimal split and explain how to solve it efficiently. [4 marks]
- (b) Explain the cost-complexity pruning objective:

$$C_\lambda(T) = \sum_{p=1}^{|T|} m_p Q_p(T) + \lambda |T|$$

What do the terms represent, and how is the optimal λ selected? [4 marks]

- (c) In AdaBoost, given distribution D_t over training examples and weak learner h_t with weighted error $\epsilon_t = \sum_{i=1}^m D_t(i) \mathbb{1}[h_t(\mathbf{x}_i) \neq y_i]$:
- (i) Write down the update rule for D_{t+1} and the coefficient α_t . [3 marks]
- (ii) Prove that the training error of the final classifier satisfies:

$$\frac{1}{m} \sum_{i=1}^m \mathbb{1}[H(\mathbf{x}_i) \neq y_i] \leq \prod_{t=1}^T Z_t$$

where Z_t is the normalization constant. [5 marks]

- (d) Contrast bagging and boosting in terms of: (i) how individual classifiers are trained, (ii) how predictions are combined, and (iii) their effect on bias versus variance. [4 marks]

5. (Online Learning: Experts and Linear Classifiers)

- (a) In the learning with expert advice setting, define the protocol and the regret of an algorithm A with respect to the best expert. [3 marks]
- (b) Consider the Weighted Majority algorithm with parameter $\beta \in [0, 1)$:
- Initialize $w_{1,i} = 1$ for all experts i .
 - Predict with weighted majority vote.
 - Update: $w_{t+1,i} = \beta w_{t,i}$ if expert i made a mistake.

Prove that if the algorithm makes M mistakes and expert i makes M_i mistakes, then:

$$M \leq \frac{\ln(1/\beta)}{\ln(2/(1+\beta))} M_i + \frac{\ln n}{\ln(2/(1+\beta))}$$

[8 marks]

- (c) State the Perceptron algorithm and the Perceptron convergence theorem (Novikoff's bound). [4 marks]
- (d) Consider learning k -literal monotone disjunctions over $\{0, 1\}^n$. Compare the mistake bounds of:
- Perceptron with feature map $\phi(\mathbf{x}) = (\mathbf{x}, 1)$
 - Winnow algorithm

Explain why Winnow has better dependence on n for sparse targets.

[5 marks]

6. (Graph-based Semi-supervised Learning)

Let $G = (V, E)$ be an undirected weighted graph with n vertices and weight matrix W where $w_{ij} \geq 0$. Let $L = D - W$ be the graph Laplacian where D is the diagonal degree matrix.

- (a) Prove that for any $\mathbf{u} \in \mathbb{R}^n$:

$$\mathbf{u}^\top L \mathbf{u} = \sum_{(i,j) \in E} w_{ij} (u_i - u_j)^2$$

What does this imply about the eigenvalues of L ?

[6 marks]

- (b) Define the harmonic energy minimization problem for graph-based transduction when vertices $1, \dots, \ell$ are labeled. Show that the solution $\bar{u}$ satisfies the harmonic property:

$$\bar{u}_i = \frac{\sum_{j=1}^n w_{ij} \bar{u}_j}{d_i} \quad \text{for unlabeled } i$$

[5 marks]

- (c) Define the effective resistance $r_G(p, q)$ between vertices p and q . Prove that:

$$r_G(p, q) = L_{pp}^+ + L_{qq}^+ - 2L_{pq}^+$$

where L^+ is the pseudo-inverse of L .

[7 marks]

- (d) Explain how the effective resistance relates to label propagation in the context of two clusters (cliques) connected by a single edge. [2 marks]

7. (Learning Theory)

- (a) Define PAC learnability for a hypothesis class H . State the role of the parameters ϵ and δ . [4 marks]

- (b) For a finite hypothesis class H under the realizability assumption, prove that ERM achieves:

$$\mathbb{P}_{S \sim \mathcal{D}^m} [L_{\mathcal{D}}(\text{ERM}_H(S)) > \epsilon] \leq |H|e^{-\epsilon m}$$

and derive the sample complexity bound. [8 marks]

- (c) Define the VC dimension of a hypothesis class H . Compute the VC dimension of:

(i) The class of threshold functions on $\mathbb{R}$: $H = \{h_a : h_a(x) = \mathbb{I}[x \geq a], a \in \mathbb{R}\}$ [2 marks]

(ii) The class of axis-aligned rectangles in $\mathbb{R}^2$ [3 marks]

- (d) State (without proof) how the VC dimension relates to the sample complexity of agnostic PAC learning. [3 marks]

8. (Online Learning: Weighted Average and Regret)

- (a) Consider the Weighted Average (WA) algorithm for learning with expert advice under general loss functions. Write down the algorithm with learning rate $\eta > 0$ and the exponential weight update. [3 marks]

- (b) For the entropic loss $L_{en}(y, \hat{y}) = y \ln(y/\hat{y}) + (1 - y) \ln((1 - y)/(1 - \hat{y}))$, prove that:

$$L_{WA}(S) - \min_{i \in [n]} L_i(S) \leq \frac{\ln n}{\eta}$$

You may use the identity relating the entropic loss to changes in KL divergence. [8 marks]

- (c) The Online Gradient Descent (OGD) algorithm for convex loss ℓ is defined by:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \ell(\mathbf{w}_t^\top \mathbf{x}_t, y_t) \mathbf{x}_t$$

State the regret bound for OGD when $\|\mathbf{w}_t\| \leq U$ and $\|\nabla\| \leq R$. [3 marks]

- (d) Explain the connection between OGD with hinge loss and the Perceptron algorithm. [3 marks]

- (e) Describe how the Fixed Share algorithm extends the standard experts setting to track the best sequence of experts with k shifts. What is the form of the regret bound? [3 marks]

END OF PAPER